

Lab 1 Report

Character-level GPT pretraining · Yelp sentiment classification · CycleGAN style transfer

Team 09 · Kaggle team “PairProgramming_Team_09”
Shibin Thomas · Teammate (name to be added)

Repository: <https://github.com/Shibin-1020/Gen-AI---LAB-1>
Branch with this work: [claude/zen-mccarthy-mpamlz](#)

	Shibin Thomas: headline result	Source
Task 1	val BPC 0.824, val perplexity 1.770, val accuracy 81.7%	task1_llm/shibin_thomas/metrics_report.csv
Task 2	best model BiGRU+attention: test accuracy 95.58%, macro-F1 0.9558, ROC-AUC 0.9917	task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/comparison.csv
Task 3	FID 117.71, MiFID 0.4217 → Kaggle score 59.06 (shown as −59.06)	task3_gan/shibin_thomas/submission.csv

Team ownership statement

Shibin Thomas built Task 1 character-level GPT (gpt_char_v1), Task 2 baseline mean-pool, TextCNN and BiGRU+attention classifiers, Task 3 CycleGAN (v1 submitted, v2 recipe), the shared TinyStories download and the shared metric modules (task1_llm/shared_eval, task2_sentiment/shared_eval), and the repository layout, reproducibility manifests and smoke tests. **Teammate (name to be added)** built Own Task 1 language model, own three Task 2 classifiers and own Task 3 CycleGAN (to be described by the teammate). Every member trained their own models in their own folder (task*/<member>/). Results were compared on shared data and shared metric code: the shared TinyStories copy and prompts, the official Yelp test split, and the instructor's Kaggle evaluation protocol. This report and its analyses were written jointly.

How to read this report. Each task section has:

- the comparison table (architecture, hyperparameters and every required metric, side by side);
- the joint analysis (strengths, weaknesses, limitations, next steps);
- an evidence table whose links open the exact log, plot or checkpoint behind each number;
- each member's failure or error analysis, with the real text or image snippets.

Numbers for Shibin Thomas are generated from the committed result files by report/build_report.py, so they match the repository exactly.

Repository layout and reproducibility

- **Per member, per task:** src/ (code and notebook), configs/, checkpoints/<run_id>/, outputs/<run_id>/, metrics_report.csv, results.md, failure_analysis.md.
- **Raw logs:** unedited, append-only, in reproducibility/raw_logs/<task>/<member>/<run_id>/.
- **Manifests:** reproducibility/manifests/<task>/<member>/<run_id>.json record the config and its hash, the git commit, library versions, hardware, a data fingerprint and the checkpoint SHA-256.
- **Smoke test:** one command per task reproduces the full pipeline at small scale (root README.md).
- **No personal paths, credentials or API keys** in any committed file.

Task 1 — GPT-style character-level language model (TinyStories)

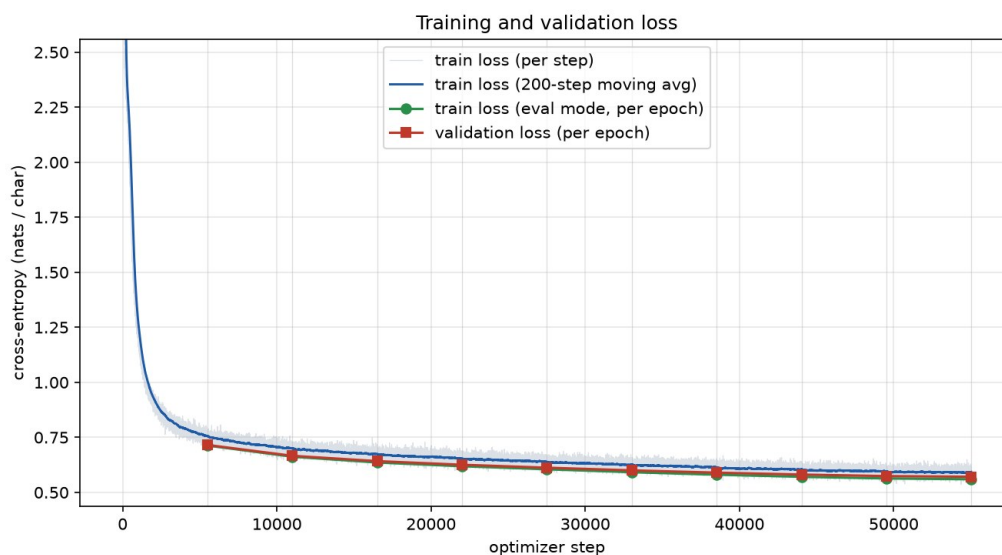
- **Shibin Thomas:** decoder-only Transformer written from scratch in `src/model.py`. LayerNorm, multi-head causal self-attention, FFN and residual blocks are all hand-written; no `nn.Transformer/nn.MultiheadAttention/scaled_dot_product_attention` (enforced by `tests/test_model.py`).
- **Data:** own 100K-train / 10K-validation split of the cleaned TinyStories pool (seed 266). Character vocabulary of 90 symbols built from the training split only.

1.1 Comparison table — architecture, hyperparameters and all Task 1 metrics

	Shibin Thomas — gpt_char_v1	teammate GPT
Architecture	6 layers, d_model 320, 8 heads (head dim 40), FFN 4x, pre-LN, GELU, learned positions, tied embeddings, context 256 chars	pending
Hyperparameters	AdamW $\beta=(0.9, 0.95)$, wd 0.1, peak LR 0.0006 → floor 6e-05, 2% warm-up + cosine, batch 64×256, clip 1.0, dropout 0.1, 10 epochs, bf16	pending
Training CE (nats/char, eval mode)	0.5594	pending
Validation CE (nats/char)	0.5711	pending
Validation perplexity	1.7701	pending
Validation BPC	0.8239	pending
Train perplexity / BPC	1.7497 / 0.8071	pending
Generalisation gap (val – train CE)	0.0116	pending
Top-1 next-char accuracy (val)	0.8173	pending
Distinct-1 / 2 / 3 (T = 0.8)	0.2277 / 0.7048 / 0.9264	pending
Repeated 4-gram rate (T = 0.8 / greedy)	0.0066 / 0.1373	pending
Gradient norm mean / p99 / max (pre-clip)	0.304 / 1.145 / 14.71	pending
Steps clipped / loss spikes / NaN steps	1.34% / 0 / 0	pending
Parameters (total / non-embedding)	7,509,120 / 7,398,400	pending
Training throughput (chars/s)	569,917	pending
Generation speed (chars/s, batch 1)	255.0	pending
Peak GPU memory (MB, allocated)	3,347	pending
Training time (min) / epochs / best epoch	28.9 / 10 / 10	pending
Hardware	NVIDIA GeForce RTX 5090	pending
Run id / best checkpoint	gpt_char_v1_20261001-153007 · checkpoints / gpt_char_v1_20261001-153007/best_model.pt (SHA-256 97647773d06f...)	pending

Metric definitions are the team's shared ones in [task1_llm/shared_eval/metrics.py](#): perplexity = $\exp(\text{CE})$, BPC = $\text{CE} / \ln 2$, Distinct-n = unique / total word n-grams of the sampled continuations. Source: [task1_llm/shibin_thomas/metrics_report.csv](#)

1.2 Training curves and per-epoch validation



Task 1 (Shibin): training loss (running, with dropout), eval-mode training CE and validation CE per epoch · source: [task1_llm/shibin_thomas/outputs/gpt_char_v1_20261001-153007/loss_curves.png](https://reproducibility/raw_logs/task1_llm/shibin_thomas/outputs/gpt_char_v1_20261001-153007/loss_curves.png)

Epoch	1	2	3	4	5	6	7	8	9	10
Val CE	0.715	0.666	0.642	0.625	0.612	0.600	0.590	0.581	0.574	0.571
Val acc.	77.4%	78.8%	79.6%	80.1%	80.5%	80.8%	81.2%	81.4%	81.6%	81.7%

Source: reproducibility/raw_logs/task1_llm/shibin_thomas/gpt_char_v1_20261001-153007/epochs.csv

1.3 Joint analysis

Strengths

- **Fluent stories from a 7.5M-parameter model trained in 29 minutes.** Validation BPC is 0.824 (81.7% next-character accuracy). Sampled stories have named characters, dialogue and simple cause and effect.
- **Clean optimisation.** Validation loss fell every epoch, with no loss spikes or NaN steps. All 738 clipped steps were in the warm-up. The generalisation gap is only 0.012 nats, as expected with about 120 training characters per parameter.
- **Sampling gives diversity.** At $T = 0.8$ the samples reach Distinct-3 0.93 with a repeated 4-gram rate of 0.7%.

Weaknesses

- **Greedy decoding loops** (repeated 4-gram rate 0.137).
- **Rare words are misspelled inconsistently**, a consequence of character-level spelling.
- **Coherence is lost beyond the 256-character context:** entities drift and stories don't end.

Limitations

- Single seed and single run per member.
- Distinct-n and repetition measure diversity, not story quality.
- Not converged: validation CE was still falling at epoch 10 (-0.003 per epoch).
- Generation speed is measured without a KV cache.

What we would try next

- More epochs, or a larger model, since the loss was still decreasing.
- Context of 512 characters, judged by long-generation coherence.
- Top-k or nucleus sampling with a repetition penalty, judged by the repeated 4-gram rate on the shared prompts.

- A KV cache for faster generation.
- A sub-word tokenizer trained from scratch, to remove spelling errors.

1.4 Evidence

Evidence	File in the repository (clickable)
Config (all hyperparameters)	task1_llm/shibin_thomas/configs/gpt_char_v1.yaml
Raw training log	reproducibility/raw_logs/task1_llm/shibin_thomas/gpt_char_v1_20261001-153007/train.log
Per-step / per-epoch logs	reproducibility/raw_logs/task1_llm/shibin_thomas/gpt_char_v1_20261001-153007
Manifest (versions, hardware, checkpoint SHA-256)	reproducibility/manifests/task1_llm/shibin_thomas/gpt_char_v1_20261001-153007.json
Loss curves / gradient norms / LR schedule	task1_llm/shibin_thomas/outputs/gpt_char_v1_20261001-153007
All 40 generated samples	task1_llm/shibin_thomas/outputs/gpt_char_v1_20261001-153007/samples.md
Best checkpoint	task1_llm/shibin_thomas/checkpoints/gpt_char_v1_20261001-153007/best_model.pt
Results, design justification	task1_llm/shibin_thomas/results.md
Executed notebook	task1_llm/shibin_thomas/src/task1_gpt_tinystories.ipynb

1.5 Failure analysis — Shibin Thomas (actual generated text)

Best checkpoint, 10 shared team prompts, 1 greedy + 3 temperature-0.8 samples each, 500 new characters. The three clearest failures (full text in `failure_analysis.md`):

Failure 1 — Repetition (greedy)

In a small house near the forest, there was a big tree. The tree was very high and the tree was very high. The tree was very high up in the tree. The tree was very high and the tree was very high.

One day, the tree saw a big tree. The tree was very high and the tree was very high. The tree was very high and the tree was very high. ...

“the tree was very” occurs 16 times; the repeated 4-gram rate of this continuation is 0.61 (average over all samples 0.04). Argmax decoding locks into a high-probability loop. **Fix:** temperature / top-k sampling or a repetition penalty, measured by the repeated 4-gram rate.

Failure 2 — Broken / inconsistent spelling (T = 0.8)

Tim was scared. He ran and ran until he came across a big lute. He saw a small tree which was very pretty. Tim wanted to take the lutter from him. He said yes, but he had to be careful.

The luter was big and boring. Tim saw that the lutte was scary and hurt. ...

A rare word is spelled four ways: lute → lutter → luter → lutte. A character model rebuilds rare words from frequent letter patterns each time. **Fix:** lower temperature / top-k, longer training, sub-word tokens.

Failure 3 — Loss of coherence / entity drift (T = 0.8)

After school, Sam wanted to go to the zoo and see all the animals.

They all walked closer and held on tight. Sam thought it was tremometer than the diamond. ... Together, Jack and Sam started to fill their bucket with food. He was so proud of their work! ... Tom and Sam continued to throw their bucket until it was time to go hom

Setting drifts (zoo → diamond → bucket → town), new companions Jack and Tom appear, and pronouns stop matching. The story is 500 characters long but the model sees only the last 256. **Fix:** longer context or a deeper model.

Teammate failure analysis: to be added from the teammate's `task1_llm/<member>/failure_analysis.md`.

Task 2 — Yelp Polarity sentiment classification (no pretrained embeddings)

- **Data:** official test split (38,000 reviews) as the shared test set; own 50K validation set (seed 266) carved from the official train split.
- **Preprocessing:** cleaning, deduplication, train/test overlap removal, lowercasing, contraction expansion, stopword removal that keeps negations, Snowball stemming, and a 30K vocabulary built from train only.
- **Embeddings:** 128-d, learned from scratch in every model.

2.1 Comparison table — every member's models, architecture, hyperparameters and all Task 2 metrics

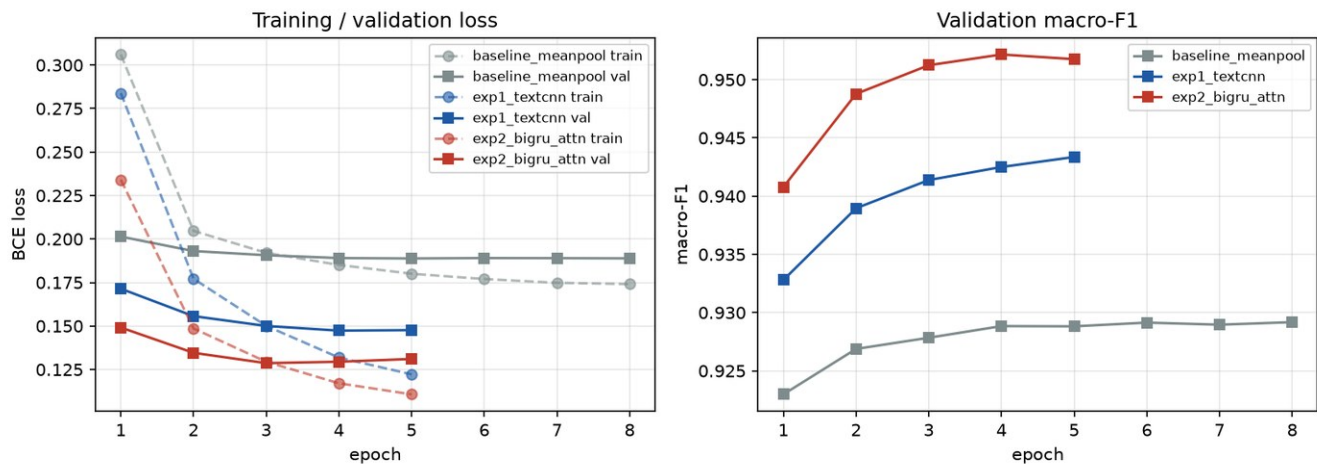
	Shibin · baseline _meanpool	Shibin · exp1_textcnn	Shibin · exp2_bigru_attn	teammate baseline	teammate experiment 1	teammate experiment 2
Architecture	Embedding 128 → masked mean → MLP 64 → 1 (bag of learned embeddings)	Embedding 128 → Conv1d widths 3/4/5 × 128 → max-over-time → dropout → linear	Embedding 128 → 2-layer BiGRU 128/dir → additive attention pooling → linear	pending	pending	pending
Hyperparameters	AdamW lr 2e-3, batch 512, dropout 0.3, ≤8 epochs, patience 2; All: 30K vocab, max_len 256 (head+tail), 3% warm-up + cosine, wd 1e-4, clip 1.0, threshold 0.5	AdamW lr 1e-3, batch 256, dropout 0.3, ≤5 epochs, patience 2	AdamW lr 1e-3, batch 256, dropout 0.3, ≤5 epochs, patience 2	pending	pending	pending
Accuracy	0.9304	0.9467	0.9558	pending	pending	pending
Accuracy 95% CI	[0.9276, 0.9330]	[0.9446, 0.9488]	[0.9537, 0.9577]	pending	pending	pending
Precision (macro)	0.9304	0.9467	0.9558	pending	pending	pending
Recall (macro)	0.9304	0.9467	0.9558	pending	pending	pending
F1 (macro)	0.9304	0.9467	0.9558	pending	pending	pending
Precision (micro)	0.9304	0.9467	0.9558	pending	pending	pending
Recall (micro)	0.9304	0.9467	0.9558	pending	pending	pending
F1 (micro)	0.9304	0.9467	0.9558	pending	pending	pending
Precision (weighted)	0.9304	0.9467	0.9558	pending	pending	pending
Recall (weighted)	0.9304	0.9467	0.9558	pending	pending	pending
F1 (weighted)	0.9304	0.9467	0.9558	pending	pending	pending
Confusion matrix [TN FP; FN TP]	[17,718 1,282; 1,363 17,637]	[17,951 1,049; 976 18,024]	[18,029 971; 710 18,290]	pending	pending	pending
ROC-AUC	0.9800	0.9883	0.9917	pending	pending	pending
PR-AUC	0.9802	0.9885	0.9920	pending	pending	pending
MCC	0.8608	0.8934	0.9116	pending	pending	pending
MCC 95% CI	[0.8553, 0.8659]	[0.8891, 0.8975]	[0.9076, 0.9155]	pending	pending	pending
Brier score	0.0520	0.0399	0.0337	pending	pending	pending
ECE (15 bins)	0.0045	0.0049	0.0110	pending	pending	pending
McNemar vs baseline (b / c, p)	- (baseline)	613 / 1,233, p=4.67e-47	557 / 1,521, p=4.65e-99	pending	pending	pending
Parameters	3,848,321	4,037,377	4,367,873	pending	pending	pending
Training time (min)	0.3545	0.9865	19.1767	pending	pending	pending
Epochs run (best)	8 (8)	5 (5)	5 (4)	pending	pending	pending
Train examples/sec	279,909	54,074	2,271	pending	pending	pending
Inference examples/sec	2,126,409	193,858	29,489	pending	pending	pending
Peak GPU memory (MB)	1,149	1,277	1,392	pending	pending	pending
Hardware	NVIDIA GeForce RTX 5090	NVIDIA GeForce RTX 5090	NVIDIA GeForce RTX 5090	pending	pending	pending
Checkpoint (SHA-256)	yelp3_20261001-182849/base_line_meanpool (725dff759...)	yelp3_20261001-182849/exp1_textcnn (b4226830a1...)	yelp3_20261001-182849/exp2_bigru_attn (2ecf44d666...)	pending	pending	pending

Official Yelp polarity test split, n = 38,000. Macro = micro = weighted because the test set is exactly balanced. Metric code: [task2_sentiment/shared_eval/metrics.py](https://github.com/shibin-thomas/task2_sentiment/blob/master/shared_eval/metrics.py). Source: [task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/comparison.csv](https://github.com/shibin-thomas/task2_sentiment/blob/master/outputs/yelp3_20261001-182849/comparison.csv)

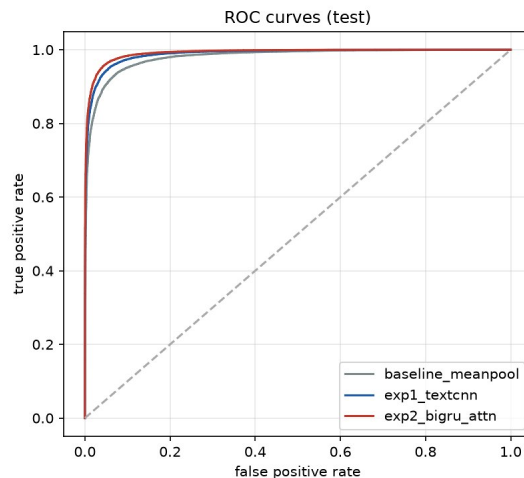
Robustness slices (error rate)

Slice (n)	baseline	TextCNN	BiGRU+attn
has negation (28,544)	7.47%	5.50%	4.48%
has contrast (22,713)	7.89%	5.94%	4.87%
no negation & no contrast (6,688)	4.93%	4.53%	4.16%
short ≤50 words (9,287)	6.96%	5.29%	5.01%
medium 51-150 words (16,910)	6.94%	5.22%	4.09%
long >150 words (11,803)	7.00%	5.52%	4.45%

Source: [task2_sentiment/shibin_thomas/metrics_report.csv](#) (slice rows) and [task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/slice_error_rates.png](#)



Task 2 (Shibin): training/validation loss and validation macro-F1 per epoch for the three models · source: [task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/training_curves.png](#)



Task 2 (Shibin): ROC curves on the official test split · source: [task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/roc_curves.png](#)

2.2 Joint analysis

Strengths

- **The ranking is significant everywhere.** BiGRU+attention > TextCNN > baseline on every discrimination metric, with non-overlapping 95% bootstrap CIs and McNemar $p \leq 1e-20$ for each pair.
- **Order-aware models fix what a bag of words cannot.** Compared with the baseline, the BiGRU removes 36% of the errors overall, 40% on negation and 38% on contrast reviews.
- **All three models are well calibrated** (ECE $\leq 1.1\%$).

- **The gains come from how the text is read, not from model size.** Parameters differ by under 0.6M; the 3.84M embedding table dominates every model.

Weaknesses

- **Speed.** The BiGRU trains 123× slower than the baseline (19 min vs 21 s) and is mildly over-confident (544 of its 1,681 errors are made with more than 90% confidence).
- **Short reviews remain the hardest slice** for the BiGRU (5.0% error).
- **Mixed-sentiment reviews dominate the confident errors.**

Limitations

- **Star-rating labels are noisy.** 5 of the 20 reviewed errors contradict their label, and 932 test reviews are wrong for all three models.
- **No pretrained knowledge** (idioms, sarcasm), as the brief requires.
- Single seed per model, and a fixed 0.5 threshold.

What we would try next

- Contrast- and negation-aware token marking, tested with McNemar on the has-contrast slice.
- max_len 512 for long reviews.
- A from-scratch Transformer encoder.
- An ensemble of the three models, with weights chosen on validation.
- Temperature scaling and several seeds.

2.3 Evidence

Evidence	File in the repository (clickable)
Configs (data + 3 models)	task2_sentiment/shibin_thomas/configs
Raw training / eval logs	reproducibility/raw_logs/task2_sentiment/shibin_thomas/yelp3_20261001-182849
Manifest (versions, hardware, checkpoint SHA-256)	reproducibility/manifests/task2_sentiment/shibin_thomas/yelp3_20261001-182849.json
All metrics, every model	task2_sentiment/shibin_thomas/metrics_report.csv
ROC / PR / reliability / slice plots	task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849
Learned-embedding neighbours (from-scratch evidence)	task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/embedding_neighbours.md
Checkpoints	task2_sentiment/shibin_thomas/checkpoints/yelp3_20261001-182849
Results, justification, comparative analysis	task2_sentiment/shibin_thomas/results.md
Executed notebook	task2_sentiment/shibin_thomas/src/task2_yelp_sentiment.ipynb

2.4 Error review — Shibin Thomas (20 misclassified test reviews, BiGRU+attention)

Selection: 5 confident false positives, 5 confident false negatives, 5 near-threshold errors and 5 from the worst slice (short reviews). Each review was read and labelled by hand.

#	True / P(pos)	Error type (manual)	Review text (excerpt)
2	neg / 1.000	Label noise	Wow love the place and everything is very clean and new! Great place to come and relax worth a try!
4	neg / 1.000	Mixed + implicit verdict	About average so far as steakhouses go. My rib eye was very tasty but a little over cooked. ... Over all I'd say they are more for show than content. If you want to have a great high end steak I'd recommend Ruth Chris...
6	pos / 0.000	Label noise (edited review)	EDIT: They really did change the service up since I last posted this. Horrible service. Used to be my favorite pizza in the city...
9	pos / 0.001	Sentiment-target confusion	Last night several parents came in with over 15 children ... The bartender expressed that it was not a place to have children running around...

#	True / P(pos)	Error type (manual)	Review text (excerpt)
13	pos / 0.498	Mixed + humour	Their alcohol selection sucks ... dreamcrushed my cocktail dreams. ... However, the burger was nice, very juicy and rare.
14	neg / 0.502	Implicit negativity / domain shift	Overly sentimental, non stop references to her kids, how blessed she is ... too many covers from elite performers...
15	pos / 0.497	Idiom misread	the pizza was to my door in 15 minutes! ... the cookie's are to die for.
16	neg / 0.999	Faint praise	Standard take out and it's cheap. Service is fast and friendly. I go here when I'm too lazy to walk to Zaw's...
18	pos / 0.002	Complex negation + star mention	despite still not digging their ordering process, their food is just too good to disrespect with a 2 star review.
19	pos / 0.002	Sarcasm / irony	For being a DUMP, should expect much more. Flys, stink, garbage, dirt ... Keepin it real dumpy!

10 of the 20 reviewed errors shown; all 20 with full text: [task2_sentiment/shibin_thomas/failure_analysis.md](#)

Error type	Count (of 20)	Model-fixable?
Mixed sentiment (aspect-level, complaints vs praise)	7	partly
Label noise (text contradicts star label)	5	no — prediction matches the text
Faint praise / implicit or comparative negativity	3	partly (world knowledge)
Complex negation, sentiment-target confusion, truncation, idiom, sarcasm	1 each	varies

Proposed testable fix: mark the clause after the last contrast word (“but”, “however”, “despite”), e.g. POST_good. Retrain the BiGRU with only this change, then compare the has-contrast slice (currently 4.87% error) and run McNemar on the same 38,000 reviews.

Teammate error review: to be added from the teammate's `task2_sentiment/<member>/failure_analysis.md`.

Task 3 — CycleGAN Monet ↔ Photo style transfer (Kaggle)

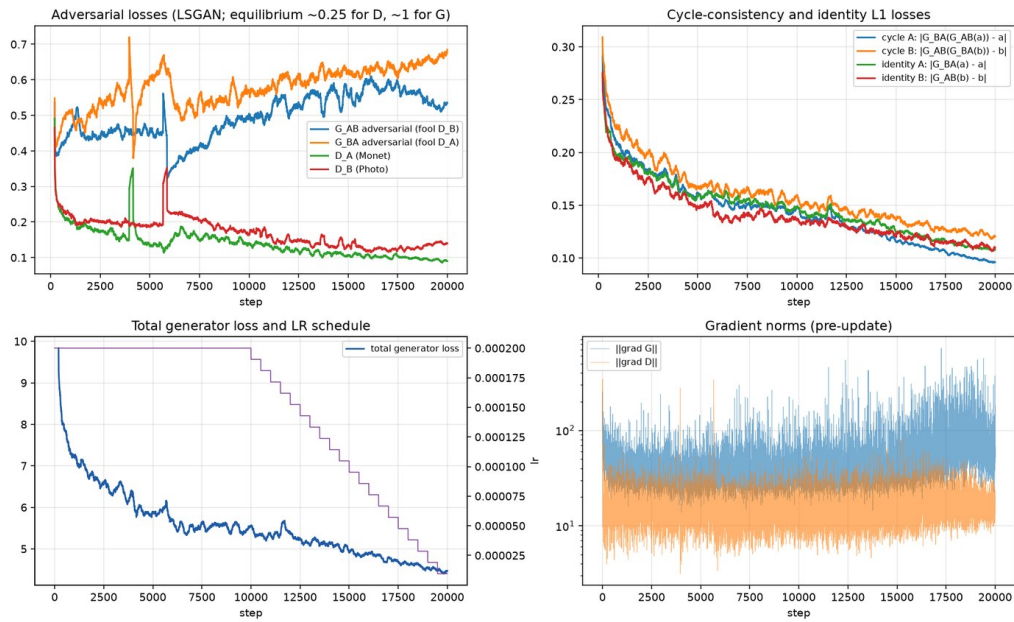
- **Shibin Thomas:** CycleGAN implemented and trained from random initialisation. Domain A = 300 Monet paintings, domain B = 7,038 photos (unpaired).
- **Kaggle protocol:** the instructor's script (Inception-v3 features, first 300 sorted images). Score = (FID + MiFID) / 2, averaged over both directions; lower is better.
- **Integrity:** the submitted images are the direct, unedited outputs of the trained generators. Pretrained networks (Inception, LPIPS) were used only to measure.

3.1 Comparison table — architecture, hyperparameters and all Task 3 metrics

	Shibin · photo→Monet (B2A)	Shibin · Monet→photo (A2B)	teammate CycleGAN
Architecture	G: ResNet-9 (c7s1-64, 2 down, 9 residual blocks, 2 resize-conv up, tanh), InstanceNorm, reflection pad, 11.38M params each · D: 70×70 PatchGAN, 2.76M each · total 28,285,832		pending
Hyperparameters	LSGAN + λ_{cyc} 10 + λ_{id} 5; Adam 2e-4 $\beta=(0.5, 0.999)$; 20 const + 20 linear-decay epochs × 2,000 pairs; batch 4; image pool 50; resize 286 → crop 256 + flip; fp32/TF32		pending
FID (instructor protocol)	118.07	117.34	pending
MiFID (script: mean paired cosine dist.)	0.4105	0.4330	pending
KID (unbiased, cubic kernel)	0.0197	0.0362	pending
Precision (k = 3, realism)	0.267	0.607	pending
Recall (k = 3, diversity)	0.640	0.320	pending
Density / coverage (k = 5)	0.221 / 0.523	0.836 / 0.757	pending
Cycle-reconstruction L1 (pixels 0-1)	0.0622	0.0531	pending
LPIPS input vs translation	0.451	0.366	pending
LPIPS input vs reconstruction	0.384	0.468	pending
Content cosine (input vs translation)	0.750	0.794	pending
Kaggle submission: FID / MiFID (avg)	117.7069 / 0.4217 → score 59.06		pending
Kaggle leaderboard (public score / rank)	pending / pending (team PairProgramming_Team_09)		pending
Final-epoch losses: G / D_A / D_B	4.458 / 0.089 / 0.139		pending
Final-epoch cycle A / B, identity A / B	0.096 / 0.120, 0.108 / 0.110		pending
Grad norm G mean / max · D mean / max	48.3 / 727 · 17.1 / 342		pending
Non-finite steps / total steps	0 / 20,000		pending
Training time / throughput / peak GPU mem.	91.2 min / 36.0 img/s / 19,922 MB		pending
Human audit (2 raters, 30 blinded samples)	pending		pending
Hardware	NVIDIA GeForce RTX 5090		pending
Run id / checkpoints (SHA-256)	cyclegan_v1_20261001-201801 · G_BA.pt e00ed38191...	G_AB.pt 4185c02dea...	pending

Source: [task3_gan/shibin_thomas/full_metrics_report.csv](#) · [task3_gan/shibin_thomas/submission.csv](#) · [task3_gan/shibin_thomas/kaggle_leaderboard.json](#) · metric code [task3_gan/shibin_thomas/src/gan_metrics.py](#)

3.2 Training behaviour and samples



Task 3 (Shibin): adversarial, cycle and identity losses, total generator loss with LR schedule, and gradient norms over 20,000 steps · source: task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/loss_curves.png

Epoch	5	10	15	20	25	30	35	40
FID B2A	153.2	145.4	147.1	128.7	130.2	119.0	123.5	121.6
FID A2B	187.7	155.2	169.5	128.6	119.7	121.8	121.3	118.1

Periodic FID on the 300 scored images (monitoring only). Source: reproducibility/raw_logs/task3_gan/shibin_thomas/cyclegan_v1_20261001-201801/epochs.csv

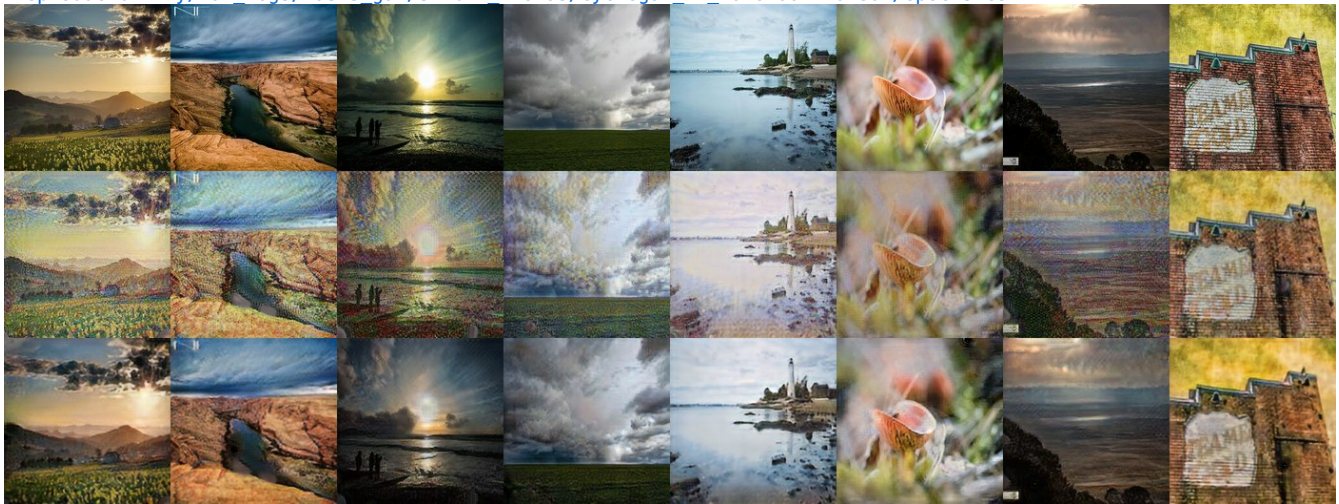


Photo → Monet: input photo (top), generated Monet (middle), reconstruction (bottom); 8 scored images · source: task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/final_samples_B2A.png



Monet → photo: input painting (top), generated photo (middle), reconstruction (bottom) · source: [task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/final_samples_A2B.png](https://github.com/shibin-thomas/task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/final_samples_A2B.png)

3.3 Joint analysis

Strengths

- **A stable from-scratch CycleGAN:** no non-finite step and no mode collapse in 20,000 steps.
- **Cycle consistency holds.** Reconstructions match the inputs (cycle L1 0.05–0.06).
- **Content and style.** Photo → Monet keeps the layout (content cosine 0.75) while moving colours to Monet's palette.
- **Exact Kaggle numbers.** They come from a re-implementation of the instructor's script, unit-tested against it.

Weaknesses

- **Photo → Monet realism is low** (precision 0.27). The cause is a regular stipple texture instead of brush strokes; dark photos are washed out.
- **Monet → photo diversity is low** (recall 0.32). Outputs stay painterly.
- **D_A over-fits the 300 paintings.** Its loss fell to 0.089, well below the 0.25 balance point.

Limitations

- FID on 300 images is noisy (± 4 between neighbouring checkpoints).
- Single seed.
- The script's MiFID (paired cosine distance between unrelated images) barely varies, so the Kaggle score is driven almost entirely by FID.
- The human audit is pending until both raters finish.

What we would try next

Implemented and ready as `configs/cyclegan_v2.yaml`:

- DiffAugment on the discriminators;
- an EMA of the generator weights;
- λ_{id} 2.5 instead of 5;
- 60 epochs instead of 40;
- epoch selection on held-out photos.

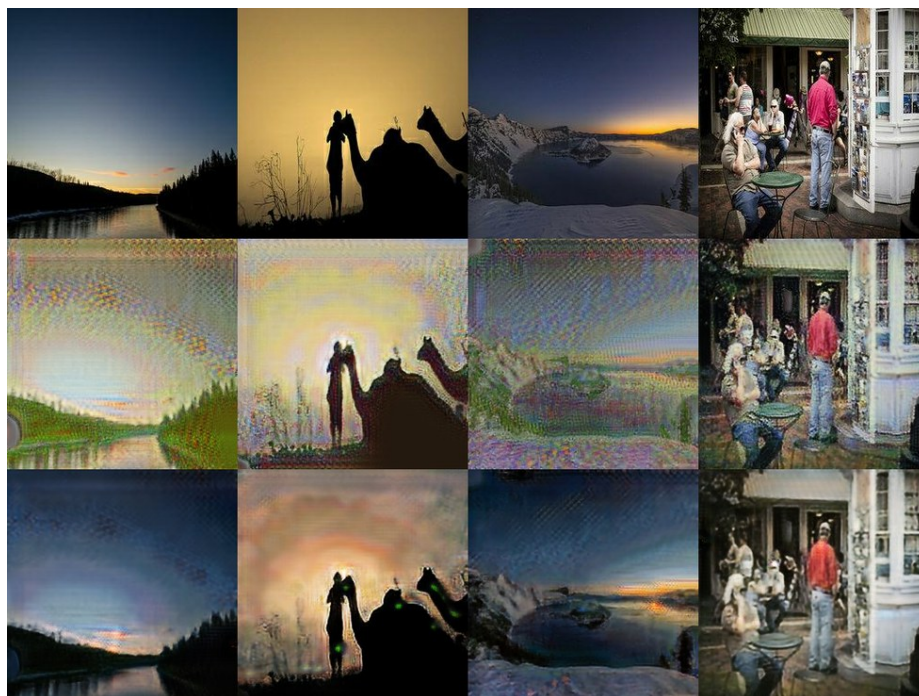
Further options: a multi-scale discriminator for stroke-level texture, and a low-pass content loss against colour inversion.

3.4 Evidence

Evidence	File in the repository (clickable)
Config	task3_gan/shibin_thomas/configs/cyclegan_v1.yaml
Raw training / eval logs (train.log, steps.csv, epochs.csv, eval.log)	reproducibility/raw_logs/task3_gan/shibin_thomas/cyclegan_v1_20261001-201801
Manifest (versions, hardware, data fingerprint)	reproducibility/manifests/task3_gan/shibin_thomas/cyclegan_v1_20261001-201801.json
Submitted predictions (300 + 300 JPGs)	task3_gan/shibin_thomas/outputs
Kaggle file / leaderboard record	task3_gan/shibin_thomas/submission.csv
Plots, sample grids, per-epoch samples	task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801
Generator checkpoints	task3_gan/shibin_thomas/checkpoints/cyclegan_v1_20261001-201801
Blinded human-audit pack	task3_gan/shibin_thomas/outputs/human_audit
Results, design justification, v2 rationale	task3_gan/shibin_thomas/results.md
Executed notebook	task3_gan/shibin_thomas/src/task3_cyclegan.ipynb

3.5 Failure analysis — Shibin Thomas (actual generated images)

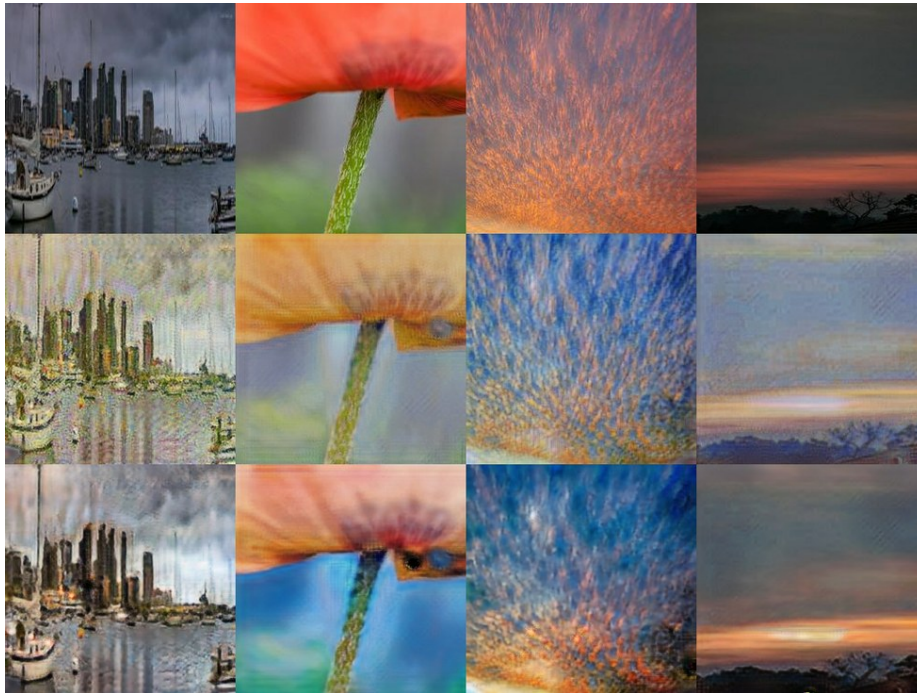
Images were ranked automatically by three criteria: highest cycle error, lowest content cosine, and largest distance to any real target image. The grids were then reviewed by hand. Each grid column shows the input (top), the translation (middle) and the reconstruction (bottom).



Failure 1 (photo → Monet): dark / night photos are washed out into bright pastel 'confetti' texture. These are the least realistic outputs (NN distance 19.5–20.3). · source:

[task3_gan/shibin_thomas/outputs/cyclegan_v1_20261001-201801/failure_B2A_nn_dist.png](#)

Cause: the 300 Monets contain almost no night scenes, so G_BA brightens every image and adds brush-like noise. **Fix:** DiffAugment colour on D_A (v2), measured by precision and NN distance on the darkest photos.



Failure 2 (photo → Monet): vivid reds and oranges are recoloured (the orange sky turns blue), yet the reconstruction recovers them: the generator hides the original colour in a faint signal (steganography). Highest cycle L1, 0.13–0.17. · source: task3_gan/shibin_thomas/outputs/cyclelegan_v1_20261001-201801/failure_B2A_cycle_l1.png

Cause: Monet's palette prior, combined with the cycle loss being satisfiable by a hidden signal; green dots in reconstructions are its visible trace. **Fix:** lower λ_{id} or add a colour term, measured by the hue shift and the cycle L1 of these images.



Failure 4 (Monet → photo): foggy, low-contrast paintings become saturated 'sunsets' or turquoise storms, and the poplar tree is lost. Lowest content cosine, 0.52–0.55. · source: task3_gan/shibin_thomas/outputs/cyclelegan_v1_20261001-201801/failure_A2B_content_cos.png

Also seen: a regular stipple texture instead of brush strokes (Failure 3; the PatchGAN judges only local statistics) and Monet → photo outputs that keep brush strokes and signatures (Failure 5). Full analysis of

all five failures: failure_analysis.md.

[task3_gan/shibin_thomas/failure_analysis.md](#)

Teammate failure analysis: to be added from the teammate's `task3_gan/<member>/failure_analysis.md`.

Pre-submission checklist

Item	Status	Where to verify
Every member's folder exists under all 3 tasks (code, config, logs, results.md)	Shibin: done · teammate: pending	task1_llm/, task2_sentiment/, task3_gan/
No two members share the same architecture + hyperparameters	to confirm when teammate rows are filled	comparison tables §1.1, §2.1, §3.1
Task 1: no prebuilt Transformer / attention modules	done (enforced by a unit test)	task1_llm/shibin_thomas/tests/test_model.py
Task 2: no pretrained embeddings / LMs; all 3 models trained per member	Shibin: done (embeddings random init, neighbours shown)	task2_sentiment/shibin_thomas/outputs/yelp3_20261001-182849/embedding_neighbours.md
All required metrics reported for every model (Tasks 1–3)	Shibin: done · teammate: pending	metrics_report.csv per member and task
Task 3 Kaggle submission made and leaderboard rank recorded	submission made; rank pending	task3_gan/shibin_thomas/kaggle_leaderboard.json
Comparison tables (architecture + hyperparameters + metrics) for all 3 tasks	done	§1.1, §2.1, §3.1
Raw logs and manifests committed and untouched	done	reproducibility/

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